

Samuel Liu

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WORK EXPERIENCE

Digital Extremes

Software Engineer

Mar 2025 - Present

- Engineered the core gameplay mechanics for fishing minigame within 'Soulframe', architecting a custom physics simulation to support realistic force-based motion and collision response, prominently showcased during 'TennoCon 2026', a live event with 500k+ concurrent livestream viewers and 4000+ in-person attendees
- Owned the end-to-end development of numerous shipped gameplay features, including an open-world castle 'siege' gameplay event spanning more than four unique locations, rapidly iterating on player feedback and internal playtesting to update and enhance the player experience across numerous patches
- Built the full stack implementation of a timestamp storing system, including backend support and gameplay integration to flexibly manage timing data across 8+ distinct systems and use cases
- Implemented custom tooling within an internal engine alongside new gameplay systems to assist with debugging, streamline designer workflows, and facilitate more efficient QA playtesting

Software Engineer (Co-op)

Sep 2023 - Dec 2023

- Developed new gameplay events and enemy AI behaviours using C++ and Lua as part of shipped 'Whispers in the Walls' major content update for 'Warframe', leading to more dynamic and nuanced player decision-making
- Improved upon existing systems of a Live-Service video game in response to post-release community feedback, enhancing player experience and driving the average monthly concurrent player count to 65,000, the highest for the game in 6 years

Behaviour Interactive

Software Engineer

Jan 2023 - Apr 2023

- Built interactive UI elements and visuals for shipped 'Silent Hill: Ascension' project within Unreal Engine 4
- Designed a tool within Unreal Engine using Slate UI to efficiently parse and compare over 20,000 assets, improving productivity of artist team by 28%

Eden Industries

Software Engineer

Jan 2022 - Apr 2022

- Spearheaded development of four distinct boss fights for a turn-based RPG using C++ and Tiled, introducing unique mechanics that encourage strategic adaptation
- Enhanced in-house game engine for performance and runtime optimization using C++, resulting in more efficient pathfinding for overworld map navigation and improved AI logic during combat encounters

EDUCATION

University of Waterloo

Sep 2019 - Apr 2024

Bachelor of Applied Science, Computer Engineering, Honours

PROJECTS

Unity 2D Game

Dec 2020 - Present

Fallen Templar | notsamyl.itch.io/fallen-templar

- Single-handedly ideated and developed a video game using C# and Unity, implementing game mechanics, algorithms for procedurally generated dungeons, and systems for audio, graphics, and gameplay
- Built a modular AI behaviour framework using finite state machines and A* pathfinding to support the rapid development of a wide range of unique enemies and bosses

SKILLS & INTERESTS

- **Languages:** C++, C#, Java, Python, Lua, Kotlin, Rust, Assembly (RISC-V), Javascript, HTML, CSS
- **Technologies:** Unity, Unreal Engine, Android Studio, React, Git, Perforce, SQL, PHP, Node, Linux, Jira
- **Interests:** swimming, rock climbing, music, video games (Outer Wilds, Bioshock Infinite, Teamfight Tactics)